🧾 Inventory Chatbot

📌 Overview

The **Inventory Chatbot** is a Flask-based web application that allows users to interact with an inventory dataset using natural language queries. It reads data from a CSV file and provides responses based on user input.

🚀 Features

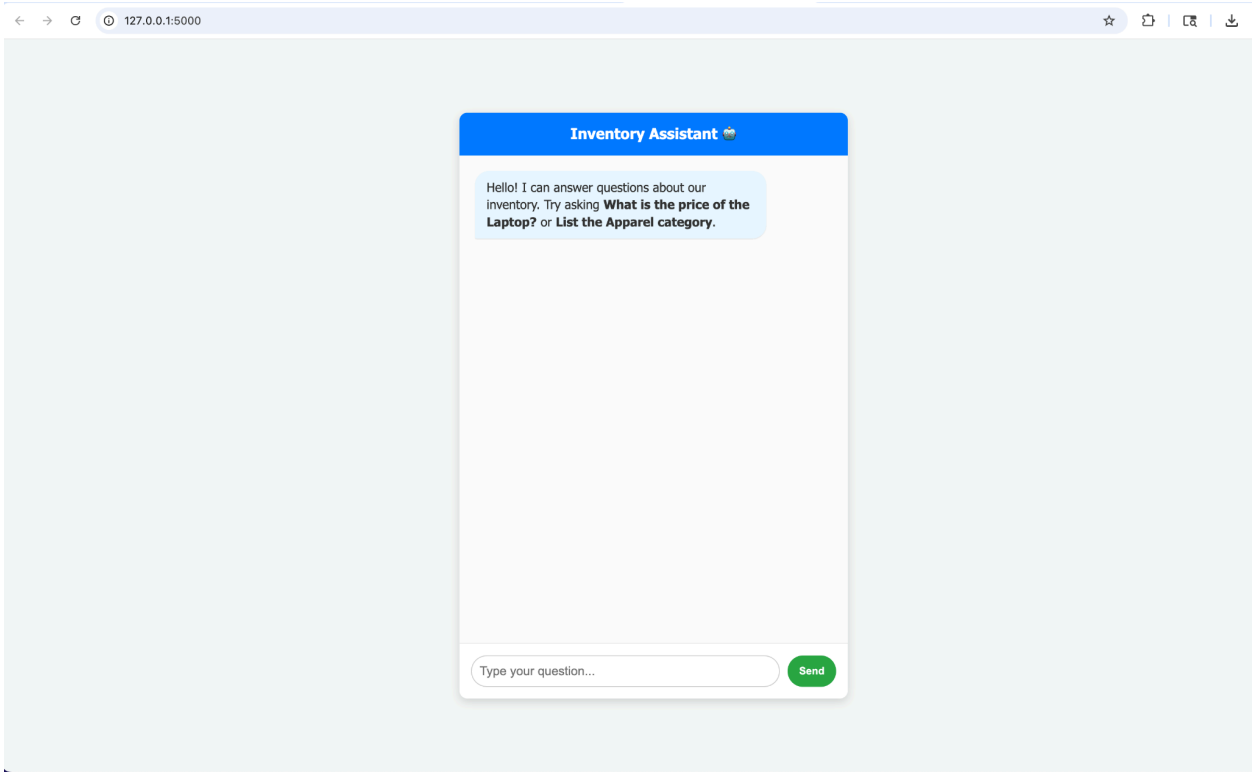
- 🔍 Query inventory using simple English
- 📦 Retrieve item details (price, category, etc.)
- 📊 Uses CSV file as database
- ⚡ Fast and lightweight (Flask-based)
- 🧠 Rule-based NLP logic (no external AI required)

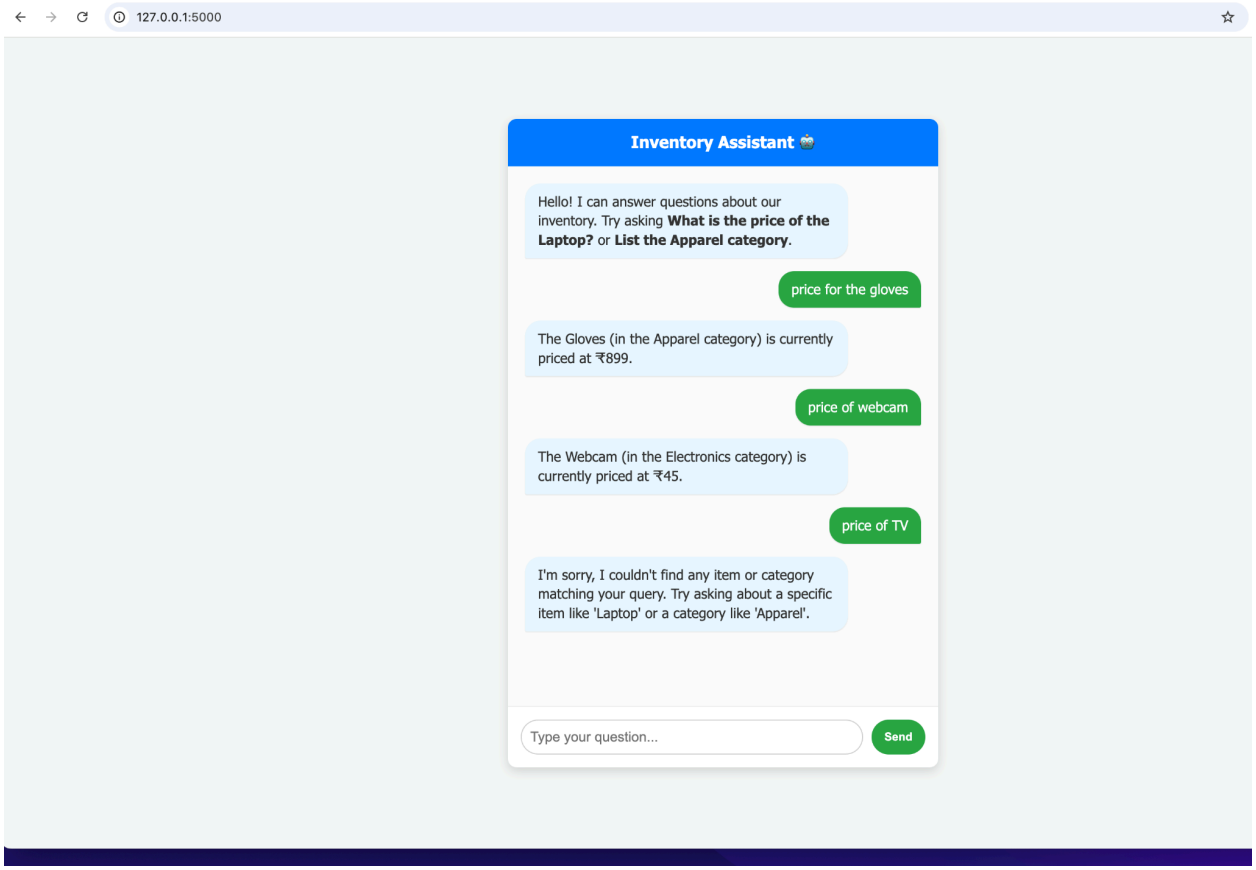
🛠️ Technologies Used

- Python
- Flask
- Pandas
- HTML (Templates)

📁 Project Structure

```
inventory Chatbot/
│── app.py          # Main Flask application
│── data.csv        # Inventory dataset
│── templates/      # HTML templates
```


127.0.0.1:5000
Inventory Assistant 🤖
Hello! I can answer questions about our inventory. Try asking **What is the price of the Laptop?** or **List the Apparel category**.
Type your question...
Send


127.0.0.1:5000
Inventory Assistant 🤖
Hello! I can answer questions about our inventory. Try asking **What is the price of the Laptop?** or **List the Apparel category**.
price for the gloves
The Gloves (in the Apparel category) is currently priced at ₹899.
price of webcam
The Webcam (in the Electronics category) is currently priced at ₹45.
price of TV
I'm sorry, I couldn't find any item or category matching your query. Try asking about a specific item like 'Laptop' or a category like 'Apparel'.
Type your question...
Send

⚙️ Installation

1. Clone the repository:

```
git clone <your-repo-link>
cd inventory-chatbot
```

2. Create a virtual environment (optional but recommended):

```
python -m venv venv
```

3. Activate the virtual environment:

- Windows:

```
venv\Scripts\activate
```

- Mac/Linux:

```
source venv/bin/activate
```

4. Install dependencies:

```
pip install flask pandas
```

▶️ Running the Application

Run the Flask app:

```
python app.py
```

Open your browser and go to:

http://127.0.0.1:5000/